

DDA3020 Machine Learning

Tutorial: K-means

April 2026

Instructions

This exercise sheet covers the main ideas of **K-means clustering** as presented in Lecture 15. You should understand the basic K-means algorithm (the assignment step and the refitting step), the distortion objective J that is being minimized, why the algorithm is guaranteed to converge and why it may get stuck at a local minimum, and how different initializations affect the final result.

Useful Formulas

K-means Objective (Distortion Measure)

Given a data set $\{\mathbf{x}_i\}_{i=1}^n$ with $\mathbf{x}_i \in \mathbb{R}^D$, K cluster centres $\{\mathbf{c}_k\}_{k=1}^K$, and binary assignment indicators $r_{ik} \in \{0, 1\}$ with $\sum_{k=1}^K r_{ik} = 1$, the K-means objective is:

$$J(\mathbf{c}, \mathbf{r}) = \sum_{i=1}^n \sum_{k=1}^K r_{ik} \|\mathbf{x}_i - \mathbf{c}_k\|^2$$

Assignment Step

Given fixed centres $\{\mathbf{c}_k\}$, for each data point $\mathbf{x}_i$, find the optimal assignment:

$$k^* = \arg \min_{1 \leq k \leq K} \|\mathbf{x}_i - \mathbf{c}_k\|^2, \quad r_{ik^*} = 1 \quad (\text{and } r_{ik} = 0 \text{ for } k \neq k^*).$$

Refitting Step

Given fixed assignments $\{r_{ik}\}$, update each centre by setting $\partial J / \partial \mathbf{c}_k = 0$:

$$\sum_{i=1}^n 2 r_{ik} (\mathbf{c}_k - \mathbf{x}_i) = 0 \quad \implies \quad \mathbf{c}_k = \frac{\sum_{i=1}^n r_{ik} \mathbf{x}_i}{\sum_{i=1}^n r_{ik}}$$

where $\sum_{i=1}^n r_{ik}$ is the number of points assigned to cluster k , and $\sum_{i=1}^n r_{ik} \mathbf{x}_i$ is the sum of all points in cluster k .

Notation reminder. Throughout this book we use the notation of the lecture slides:

- $\{\mathbf{x}_i\}_{i=1}^n$: the data set of n points, $\mathbf{x}_i \in \mathbb{R}^D$.
- $\{\mathbf{c}_k\}_{k=1}^K$ (sometimes written $\{\boldsymbol{\mu}_k\}$): the K cluster centres.
- $r_{ik} \in \{0, 1\}$ with $\sum_{k=1}^K r_{ik} = 1$: hard-assignment indicator ($r_{ik} = 1$ iff $\mathbf{x}_i$ is in cluster k). Some sources write γ_{ij} for the same quantity.
- $J(\mathbf{c}, \mathbf{r}) = \sum_{i=1}^n \sum_{k=1}^K r_{ik} \|\mathbf{x}_i - \mathbf{c}_k\|^2$: the within-cluster sum-of-squares objective (the *distortion measure*).

Additional Exercises

Q1. Suppose we have the following one-dimensional points:

$$x_1 = 0, \quad x_2 = 2, \quad x_3 = 3, \quad x_4 = 8, \quad x_5 = 10.$$

Run K-means with $K = 2$ until convergence, using the initial centres

$$c_1^{(0)} = -1, \quad c_2^{(0)} = 5.$$

Use squared Euclidean distance. If a point is tied between two clusters, assign it to the cluster with the smaller index.

- Perform the assignment and refitting steps until convergence.
- How many iterations are needed? Do not double-count the final unchanged assignment.
- What are the final assignments and final centres?
- Compute the final K-means objective

$$J = \sum_{i=1}^5 \|\mathbf{x}_i - \mathbf{c}_{z_i}\|^2.$$

- In general, does K-means always converge to the global minimum?

Q2. Consider the following two-dimensional data points:

$$\mathbf{x}_1 = (0, 0), \quad \mathbf{x}_2 = (0, 2), \quad \mathbf{x}_3 = (2, 0), \quad \mathbf{x}_4 = (8, 8), \quad \mathbf{x}_5 = (8, 10), \quad \mathbf{x}_6 = (10, 8).$$

Run one full iteration of K-means with $K = 2$, using the initial centres

$$\mathbf{c}_1^{(0)} = (0, 0), \quad \mathbf{c}_2^{(0)} = (8, 8).$$

Use squared Euclidean distance.

- Assign each point to its nearest centre.
- Write the assignment matrix $\mathbf{r} \in \{0, 1\}^{6 \times 2}$, where $r_{ik} = 1$ means $\mathbf{x}_i$ is assigned to cluster k .
- Refit the two cluster centres by taking the mean of the assigned points.

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- (d) Compute the K-means objective after the assignment step but before refitting:

$$J_{\text{old}} = \sum_{i=1}^6 \sum_{k=1}^2 r_{ik} \|\mathbf{x}_i - \mathbf{c}_k^{(0)}\|^2.$$

Then compute the objective after refitting:

$$J_{\text{new}} = \sum_{i=1}^6 \sum_{k=1}^2 r_{ik} \|\mathbf{x}_i - \mathbf{c}_k^{(1)}\|^2.$$

- (e) Does the objective decrease after the refitting step?

Q3. Consider four two-dimensional data points:

$$(-3, 0), \quad (-3, 1), \quad (3, 0), \quad (3, 1).$$

Run K-means with $K = 2$ using squared Euclidean distance.

- (a) If the initial centres are $(-3, 0)$ and $(3, 0)$, find the final clusters, final centres, and final objective value.
- (b) If the initial centres are $(-3, 0)$ and $(-3, 1)$, find the final clusters, final centres, and final objective value.
- (c) Which final clustering is better according to the K-means objective?
- (d) What does this example show about random initialization, local minima, and multiple restarts?

Q4. A 512×512 RGB image stores each pixel with three 8-bit color channels, so each pixel uses 24 bits. We compress the image by treating each pixel as a point in $\mathbb{R}^3$, running K-means with $K = 16$, and replacing each pixel by its nearest centroid color.

- (a) How many bits are needed to store the cluster index of one pixel after compression?
- (b) Ignoring the cost of storing the 16 centroid colors, what is the approximate compression factor?
- (c) If the codebook of 16 RGB centroids must also be stored, how many extra bits are required?
- (d) Is this codebook cost important compared with the original image size?

Q5. (a) For a sample $\mathbf{x}_i$, the silhouette coefficient is

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)},$$

where a_i is the mean distance from $\mathbf{x}_i$ to points in its own cluster, and b_i is the smallest mean distance from $\mathbf{x}_i$ to points in another cluster. What is the range of s_i , and what do values close to 1, 0, and -1 mean?

- (b) In Exercise 1, the final clustering is $C_1 = \{0, 2, 3\}$ and $C_2 = \{8, 10\}$. Compute the silhouette coefficient of the point $x = 2$, using ordinary Euclidean distance.

(c) Let the ground-truth partition of six objects be

$$X = \{\{1, 2, 3\}, \{4, 5, 6\}\},$$

and let a clustering algorithm return

$$Y = \{\{1, 2\}, \{3, 4\}, \{5, 6\}\}.$$

Compute the Rand index.

(d) Build the contingency table for X and Y , and compute the adjusted Rand index:

$$\text{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}.$$

Q6. Consider the K-means algorithm discussed in the lecture. Show that, as a consequence of (i) there being a finite number of possible assignments for the set of discrete indicator variables r_{ik} , and (ii) for each such assignment there being a unique optimum for the centres $\{c_k\}$, the K-means algorithm must converge after a finite number of iterations.

Q7. (*This exercise goes slightly beyond the main slide content. It is an optional extension inspired by the variants topic on Slide 35 and is included here for students interested in a deeper understanding.*)

Suppose we replace the squared Euclidean distance in the distortion measure with the ℓ_1 norm:

$$J'(\gamma, \mu_1, \dots, \mu_K) = \sum_{i=1}^n \sum_{j=1}^K \gamma_{ij} \|\mathbf{x}_i - \mu_j\|_1.$$

- (a) Derive the steps of the new K-means formulation for this ℓ_1 objective. (The answer may not be unique.)
- (b) If your data contain *outliers*, which version of K-means would you use — the ℓ_1 version or the original squared-Euclidean one? Justify your answer.

Q8. There is a set S consisting of 6 points in the plane shown below:

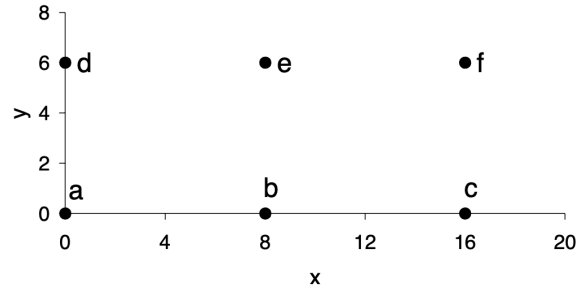
$$a = (0, 0), \quad b = (8, 0), \quad c = (16, 0), \quad d = (0, 6), \quad e = (8, 6), \quad f = (16, 6).$$

We run the K-means algorithm on those points with $K = 3$. The algorithm uses the Euclidean distance metric (i.e. the straight-line distance between two points) to assign each point to its nearest centroid. Ties are broken in favor of the centroid to the left/down. Two definitions:

- A **K -starting configuration** is a subset of K starting points from S that form the initial centroids (e.g. $\{a, b, c\}$).
- A **K -partition** is a partition of S into K non-empty subsets (e.g. $\{a, b, e\}, \{c, d\}, \{f\}$ is a 3-partition).

Clearly any K -partition induces a set of K centroids in the natural manner. A K -partition is called **stable** if a repetition of the K-means iteration with the induced centroids leaves it unchanged.

- (a) How many 3-starting configurations are there? (Remember: a 3-starting configuration is just a subset, of size 3, of the six data points.)



- (b) Fill in the following table. For each candidate 3-partition, decide whether it is stable; if so, give one example of a 3-starting configuration that can arrive at the partition after 0 or more iterations of K-means (or write “none” if no such starting configuration exists); and give the total number of starting configurations that reach it.

3-partition	Stable?	Example starting config.	# starts
$\{a, b, e\}, \{c, d\}, \{f\}$			
$\{a, b\}, \{d, e\}, \{c, f\}$			
$\{a, d\}, \{b, e\}, \{c, f\}$			
$\{a\}, \{d\}, \{b, c, e, f\}$			
$\{a, b\}, \{d\}, \{c, e, f\}$			
$\{a, b, d\}, \{c\}, \{e, f\}$			

Q9. For each of the following statements, decide whether it is **True** or **False**, and justify your answer in one or two sentences.

- For a particular data set and a particular K , K-means always produces the same result if the initialized centres are the same. (Assume there is no tie when assigning the clusters.)
- K-means can always converge to the global optimum.
- K-means is not sensitive to outliers.
- In K-means, the cost J always drops (or stays the same) after one complete update step (i.e. one assignment step followed by one refitting step).

Q10. Consider a data set with seven points $\{x_1, \dots, x_7\}$. The pairwise distances between all points are given in the following table.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7
x_1	0	5	3	1	6	2	3
x_2	5	0	4	6	1	7	8
x_3	3	4	0	4	3	5	6
x_4	1	6	4	0	7	1	2
x_5	6	1	3	7	0	8	9
x_6	2	7	5	1	8	0	1
x_7	3	8	6	2	9	1	0

Assume $K = 2$, and the cluster centres are initialized to x_3 and x_6 . After one iteration of K-means (i.e. one assignment step using the distances above), which of the following shows the two clusters formed? Circle the correct option.

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- (a) $\{x_1, x_2, x_3, x_4\}, \{x_5, x_6, x_7\}$
- (b) $\{x_2, x_3, x_5\}, \{x_1, x_4, x_6, x_7\}$
- (c) $\{x_1, x_2, x_3, x_5\}, \{x_4, x_6, x_7\}$
- (d) $\{x_2, x_3, x_4, x_7\}, \{x_1, x_5, x_6\}$

Solutions

Q1. Answer:

(a) Iteration 1, with centres -1 and 5 :

$$C_1 = \{0, 2\}, \quad C_2 = \{3, 8, 10\}.$$

The point $x = 2$ is tied between the two centres, so we assign it to cluster 1. The refitted centres are

$$c_1^{(1)} = \frac{0+2}{2} = 1, \quad c_2^{(1)} = \frac{3+8+10}{3} = 7.$$

Iteration 2, with centres 1 and 7 :

$$C_1 = \{0, 2, 3\}, \quad C_2 = \{8, 10\}.$$

The refitted centres are

$$c_1^{(2)} = \frac{0+2+3}{3} = \frac{5}{3}, \quad c_2^{(2)} = \frac{8+10}{2} = 9.$$

Iteration 3 gives the same assignment, so the algorithm has converged.

(b) The algorithm converges in 2 iterations, since the next assignment is unchanged.

(c) The final assignments are

$$z_1 = z_2 = z_3 = 1, \quad z_4 = z_5 = 2.$$

The final centres are

$$c_1 = \frac{5}{3}, \quad c_2 = 9.$$

(d) The final objective is

$$\begin{aligned} J &= \left(0 - \frac{5}{3}\right)^2 + \left(2 - \frac{5}{3}\right)^2 + \left(3 - \frac{5}{3}\right)^2 + (8 - 9)^2 + (10 - 9)^2 \\ &= \frac{25}{9} + \frac{1}{9} + \frac{16}{9} + 1 + 1 = \frac{20}{3}. \end{aligned}$$

(e) No. K-means monotonically decreases the objective for a fixed initialization, but it may converge to a local minimum rather than the global minimum.

Q2. Answer:

(a) The first three points are closer to $(0, 0)$, and the last three points are closer to $(8, 8)$. Thus

$$C_1 = \{(0, 0), (0, 2), (2, 0)\}, \quad C_2 = \{(8, 8), (8, 10), (10, 8)\}.$$

(b) The assignment matrix is

$$\mathbf{r} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}.$$

(c) The refitted centres are

$$\mathbf{c}_1^{(1)} = \frac{(0,0) + (0,2) + (2,0)}{3} = \left(\frac{2}{3}, \frac{2}{3}\right),$$

$$\mathbf{c}_2^{(1)} = \frac{(8,8) + (8,10) + (10,8)}{3} = \left(\frac{26}{3}, \frac{26}{3}\right).$$

(d) Before refitting, the objective is

$$J_{\text{old}} = 0 + 4 + 4 + 0 + 4 + 4 = 16.$$

After refitting, the first cluster contributes

$$\left\| (0,0) - \left(\frac{2}{3}, \frac{2}{3}\right) \right\|^2 + \left\| (0,2) - \left(\frac{2}{3}, \frac{2}{3}\right) \right\|^2 + \left\| (2,0) - \left(\frac{2}{3}, \frac{2}{3}\right) \right\|^2 = \frac{16}{3}.$$

The second cluster contributes the same amount, so

$$J_{\text{new}} = \frac{16}{3} + \frac{16}{3} = \frac{32}{3}.$$

(e) Yes. The objective decreases from

$$16 \quad \text{to} \quad \frac{32}{3} \approx 10.67.$$

Q3. Answer:

(a) With initial centres $(-3,0)$ and $(3,0)$, the stable clustering is the vertical split:

$$C_1 = \{(-3,0), (-3,1)\}, \quad C_2 = \{(3,0), (3,1)\}.$$

The centres are

$$\mathbf{c}_1 = (-3, 0.5), \quad \mathbf{c}_2 = (3, 0.5).$$

Each point has squared distance 0.25 to its centre, so

$$J = 4(0.25) = 1.$$

(b) With initial centres $(-3,0)$ and $(-3,1)$, the stable clustering is the horizontal split:

$$C_1 = \{(-3,0), (3,0)\}, \quad C_2 = \{(-3,1), (3,1)\}.$$

The centres are

$$\mathbf{c}_1 = (0, 0), \quad \mathbf{c}_2 = (0, 1).$$

Each point has squared distance 9 to its centre, so

$$J = 4(9) = 36.$$

(c) The first clustering is much better, since $1 < 36$.

(d) The example shows that different initializations can produce very different final objectives. Running K-means several times and selecting the run with the lowest objective reduces the risk of accepting a poor local minimum.

Q4. Answer:

(a) Since $K = 16 = 2^4$, one pixel needs 4 bits to store its cluster index.

(b) Originally each pixel uses 24 bits. After replacing RGB values by a 16-color index, each pixel uses 4 bits. Ignoring the codebook, the compressed representation uses

$$\frac{4}{24} = \frac{1}{6}$$

of the original per-pixel storage, so the image is compressed by about a factor of 6.

(c) Each centroid has three 8-bit channels, so the codebook costs

$$16 \cdot 3 \cdot 8 = 384$$

bits.

(d) The original image uses

$$512 \cdot 512 \cdot 24 = 6,291,456$$

bits. The 384-bit codebook is negligible compared with the original image size.

Q5. Answer:

(a) The silhouette coefficient satisfies

$$-1 \leq s_i \leq 1.$$

A value close to 1 means the point is much closer to its own cluster than to other clusters. A value close to 0 means it lies near a boundary between clusters. A value close to -1 means it may be assigned to the wrong cluster.

(b) For $x = 2$, the within-cluster mean distance is

$$a = \frac{|2 - 0| + |2 - 3|}{2} = \frac{3}{2}.$$

The mean distance to the other cluster is

$$b = \frac{|2 - 8| + |2 - 10|}{2} = 7.$$

Therefore

$$s = \frac{b - a}{\max(a, b)} = \frac{7 - \frac{3}{2}}{7} = \frac{11}{14} \approx 0.786.$$

(c) Count all $\binom{6}{2} = 15$ unordered pairs. The pairs in the same subset in both partitions are

$$(1, 2), \quad (5, 6),$$

so $a = 2$. The pairs in different subsets in both partitions are $b = 8$. Thus

$$\text{RI} = \frac{a + b}{\binom{6}{2}} = \frac{2 + 8}{15} = \frac{2}{3}.$$

(d) The contingency table is

	$\{1, 2\}$	$\{3, 4\}$	$\{5, 6\}$	a_i
$\{1, 2, 3\}$	2	1	0	3
$\{4, 5, 6\}$	0	1	2	3
b_j	2	2	2	6

Hence

$$\sum_{ij} \binom{n_{ij}}{2} = 2, \quad \sum_i \binom{a_i}{2} = 6, \quad \sum_j \binom{b_j}{2} = 3, \quad \binom{n}{2} = 15.$$

Therefore

$$\text{ARI} = \frac{2 - \frac{6 \cdot 3}{15}}{\frac{1}{2}(6 + 3) - \frac{6 \cdot 3}{15}} = \frac{2 - \frac{6}{5}}{\frac{9}{2} - \frac{6}{5}} = \frac{8}{33} \approx 0.242.$$

Q6. Answer:

Since both the assignment step and the refitting step minimize the distortion measure J , the algorithm will never change from a particular assignment of data points to prototypes, unless the new assignment has a lower value for J .

Since there is a finite number of possible assignments, each with a corresponding unique minimum of J with respect to the prototypes $\{\mu_k\}$, the K-means algorithm will converge after a finite number of steps, when no re-assignment of data points to prototypes will result in a decrease of J . When no re-assignment takes place, there also will not be any change in $\{\mu_k\}$.

Q7. Answer:

(a) **Derivation of the ℓ_1 K-means steps.** The structure of the algorithm is unchanged; only the two update rules differ.

- Initialize $\mu_1, \dots, \mu_K$.
- While J' is decreasing, repeat:

(1) *Assignment.* Determine γ (breaking ties arbitrarily):

$$\gamma_{ij} = \begin{cases} 1, & \|\mathbf{x}_i - \mu_j\|_1 \leq \|\mathbf{x}_i - \mu_{j'}\|_1 \quad \forall j' \\ 0, & \text{otherwise.} \end{cases}$$

(2) *Refitting.* Recompute μ_j using the updated γ . Remove j from the set of clusters if $\sum_{i=1}^n \gamma_{ij} = 0$; otherwise

$$\mu_j = \text{median}(\{\mathbf{x}_i : \gamma_{ij} = 1\}).$$

Why the median? The goal is to minimize $\sum_{i: \gamma_{ij}=1} \|\mathbf{x}_i - \mu_j\|_1$ for the j -th cluster. Picking the μ_j that minimizes this sum gives our new centroid. Consider the derivative of this objective with respect to μ_j (component-wise). Each of the ℓ_1 terms contributes either +1 or -1 depending on whether the component of the point is greater or smaller than the corresponding component of μ_j . This derivative is 0 when the number of +1's and -1's cancel — i.e. when there are an equal number of points on either side of the centroid. The choice of centroid that achieves this is the (component-wise) median of the points in the cluster. The solution may not be unique for a given cluster.

(b) **Outlier robustness.** We might prefer the ℓ_1 objective for noisy data because the *geometric median* is robust to outliers. The median only cares about picking the middle point in the ordering; it does not care about the distance to the furthest points, only the number of points on each side. In contrast, the mean (which is what the squared-Euclidean K-means uses) is pulled strongly toward outliers because each large residual is *squared*. So if the data contains outliers, the ℓ_1 version is the safer choice.

Q8. Answer:

(a) A 3-starting configuration is a size-3 subset of the six data points, so the number of such configurations is

$$\binom{6}{3} = 20.$$

(b) The completed table is shown below.

3-partition	Stable?	Example starting config.	# starts
$\{a, b, e\}, \{c, d\}, \{f\}$	N	none	0
$\{a, b\}, \{d, e\}, \{c, f\}$	Y	$\{b, c, e\}$	4
$\{a, d\}, \{b, e\}, \{c, f\}$	Y	$\{a, b, c\}$	8
$\{a\}, \{d\}, \{b, c, e, f\}$	Y	$\{a, b, d\}$	2
$\{a, b\}, \{d\}, \{c, e, f\}$	Y	none	0
$\{a, b, d\}, \{c\}, \{e, f\}$	Y	$\{a, c, f\}$	1

Remark. A partition is stable exactly when applying one iteration of K-means to its induced centroids returns the same partition. The first row is not stable: given the centroids induced by $\{a, b, e\}, \{c, d\}, \{f\}$, a re-assignment of the six points to the nearest of those three centroids does not reproduce the same partition. For the fifth row, although the partition $\{a, b\}, \{d\}, \{c, e, f\}$ is itself stable, no starting configuration (out of the 20) can reach it through 0 or more iterations of K-means.

Q9. Answer:

(a) **True.** Every time you are computing the completely same distances, so the result is the same. The K-means algorithm is entirely deterministic once the initial centres are fixed (and assuming no ties in the assignment step).

(b) **False.** It depends on the initialization. Random initialization could possibly lead to a local optimum, not the global one. This is the reason Slide 32 recommends running K-means with multiple random initializations and picking the run with the lowest objective value.

(c) **False.** K-means is quite sensitive to outliers, since it computes the cluster centre based on the *mean* value of all data points in the cluster. A single outlier that is far from the rest can pull the mean significantly, distorting the resulting cluster.

(d) **True.** Each assignment step finds the global minimum of J with respect to $\mathbf{r}$ for the current centres, and each refitting step finds the global minimum of J with respect to $\{\mu_k\}$ for the current assignments. Therefore J is non-increasing across every complete update step.

Q10. Answer:

With the initial centres at x_3 and x_6 , we assign each point to its closer centre:

Point	dist to x_3	dist to x_6	Assigned to
x_1	3	2	x_6
x_2	4	7	x_3
x_3	0	5	x_3
x_4	4	1	x_6
x_5	3	8	x_3
x_6	5	0	x_6
x_7	6	1	x_6

The two clusters after the first assignment step are therefore

$$\{x_2, x_3, x_5\} \quad \text{and} \quad \{x_1, x_4, x_6, x_7\}.$$

The answer is **(b)**.